ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Database Management System

Assignment 6

CAB BOOKING SYSTEM

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Course	Database Management System
Programme	BCA AI/ML
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Recreating the tables with all constraints

```
DROP TABLE Payment;
DROP TABLE Booking;
DROP TABLE Car;
DROP TABLE Driver;
DROP TABLE Customer;
```

```
CREATE TABLE Customer (
    Customer_ID    VARCHAR(10)  NOT NULL,
    Customer_Name  VARCHAR(50)  NOT NULL,
    Phone          VARCHAR(15)  NOT NULL,
    Email          VARCHAR(50),
    Address        VARCHAR(100) NOT NULL DEFAULT 'Not Provided',
    CONSTRAINT pk_customer PRIMARY KEY (Customer_ID),
    CONSTRAINT uq_customer_phone UNIQUE (Phone),
    CONSTRAINT uq_customer_email UNIQUE (Email),
    CONSTRAINT chk_customer_phone CHECK (LENGTH(Phone) = 10)
);

CREATE TABLE Driver (
    Driver_ID      VARCHAR(10)  NOT NULL,
    Driver_Name    VARCHAR(50)  NOT NULL,
    Phone          VARCHAR(15)  NOT NULL,
    License_No     VARCHAR(20)  NOT NULL,
    Availability   VARCHAR(15)  NOT NULL DEFAULT 'Available',
    CONSTRAINT pk_driver PRIMARY KEY (Driver_ID),
    CONSTRAINT uq_driver_phone UNIQUE (Phone),
    CONSTRAINT uq_driver_license UNIQUE (License_No),
    CONSTRAINT chk_driver_avail
        CHECK (Availability IN ('Available', 'Busy', 'Offline'))
);

CREATE TABLE Car (
    Car_ID         VARCHAR(10)  NOT NULL,
    Car_Number     VARCHAR(15)  NOT NULL,
    Car_Model      VARCHAR(30)  NOT NULL,
    Car_Type       VARCHAR(20)  NOT NULL DEFAULT 'Hatchback',
    Fare_Per_KM    DECIMAL(6,2) NOT NULL,
    Driver_ID      VARCHAR(10),
    CONSTRAINT pk_car PRIMARY KEY (Car_ID),
    CONSTRAINT uq_car_number UNIQUE (Car_Number),
    CONSTRAINT chk_car_type
        CHECK (Car_Type IN ('Hatchback', 'Sedan', 'SUV')),
    CONSTRAINT chk_car_fare CHECK (Fare_Per_KM > 0),
    CONSTRAINT fk_car_driver FOREIGN KEY (Driver_ID)
        REFERENCES Driver(Driver_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE Booking (
    Booking_ID       VARCHAR(10) NOT NULL,
    Customer_ID      VARCHAR(10) NOT NULL,
    Car_ID           VARCHAR(10) NOT NULL,
    Booking_Date     DATE        NOT NULL DEFAULT (CURRENT_DATE),
    Pickup_Location  VARCHAR(50) NOT NULL,
    Drop_Location    VARCHAR(50) NOT NULL,
    Distance_KM      INT         NOT NULL,
    Booking_Status   VARCHAR(15) NOT NULL DEFAULT 'Confirmed',
    CONSTRAINT pk_booking PRIMARY KEY (Booking_ID),
    CONSTRAINT chk_booking_distance CHECK (Distance_KM > 0),
    CONSTRAINT chk_booking_status
        CHECK (Booking_Status IN ('Confirmed', 'Completed', 'Cancelled')),
    CONSTRAINT chk_booking_places CHECK (Pickup_Location <> Drop_Location),
    CONSTRAINT fk_booking_customer FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_booking_car FOREIGN KEY (Car_ID)
        REFERENCES Car(Car_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

```
CREATE TABLE Payment (
    Payment_ID      VARCHAR(10)  NOT NULL,
    Booking_ID      VARCHAR(10)  NOT NULL,
    Amount          DECIMAL(8,2) NOT NULL,
    Payment_Date    DATE         NOT NULL,
    Payment_Method  VARCHAR(10)  NOT NULL DEFAULT 'Cash',
    Payment_Status  VARCHAR(10)  NOT NULL DEFAULT 'Pending',
    CONSTRAINT pk_payment PRIMARY KEY (Payment_ID),
    CONSTRAINT uq_payment_booking UNIQUE (Booking_ID),
    CONSTRAINT chk_payment_amount CHECK (Amount >= 0),
    CONSTRAINT chk_payment_method
        CHECK (Payment_Method IN ('Cash', 'UPI', 'Card')),
    CONSTRAINT chk_payment_status
        CHECK (Payment_Status IN ('Paid', 'Pending', 'Refunded')),
    CONSTRAINT fk_payment_booking FOREIGN KEY (Booking_ID)
        REFERENCES Booking(Booking_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

Purpose of the constraints used

Constraint	Where I used it	Why it is needed
PRIMARY KEY	Customer_ID, Driver_ID, Car_ID, Booking_ID, Payment_ID	Identifies each row uniquely. It cannot be repeated or NULL, so there are no duplicate customers, bookings, etc.
FOREIGN KEY	Car.Driver_ID, Booking.Customer_ID, Booking.Car_ID, Payment.Booking_ID	Makes sure a row can only refer to a row that really exists in the parent table. ON DELETE RESTRICT stops deleting a parent row that still has child rows. ON UPDATE CASCADE copies an ID change to the child rows.
UNIQUE	Phone and Email (Customer), Phone and License_No (Driver), Car_Number (Car), Booking_ID (Payment)	No two people can share a phone, email or licence, and no two cars can have the same number. UNIQUE on Payment.Booking_ID makes the 1:1 relationship strict, so one booking can have only one payment.
CHECK	Phone has 10 digits; Availability, Car_Type, Booking_Status, Payment_Method and Payment_Status limited to allowed values; Fare_Per_KM > 0; Distance_KM > 0; Amount >= 0; Pickup <> Drop	Stops values that make no sense, such as a negative fare, a booking status like "Maybe", or a trip that starts and ends at the same place.
DEFAULT	Address = Not Provided; Availability = Available; Car_Type = Hatchback; Booking_Date = today; Booking_Status = Confirmed; Payment_Method = Cash; Payment_Status = Pending	Fills in a sensible value automatically when none is given, which saves typing and avoids empty values.
NOT NULL	Names, phone, licence, fare, locations, distance, amount, dates	These details are compulsory, so a row cannot be saved without them.

Structure after adding the constraints

```
DESCRIBE Customer;
```

Output:

Field	Type	Null	Key	Default
Customer_ID	varchar(10)	NO	PRI	NULL

Field	Type	Null	Key	Default
Customer_Name	varchar(50)	NO		NULL
Phone	varchar(15)	NO	UNI	NULL
Email	varchar(50)	YES	UNI	NULL
Address	varchar(100)	NO		'Not Provided'

```
DESCRIBE Booking;
```

Output:

Field	Type	Null	Key	Default
Booking_ID	varchar(10)	NO	PRI	NULL
Customer_ID	varchar(10)	NO	MUL	NULL
Car_ID	varchar(10)	NO	MUL	NULL
Booking_Date	date	NO		CURRENT_DATE
Pickup_Location	varchar(50)	NO		NULL
Drop_Location	varchar(50)	NO		NULL
Distance_KM	int	NO		NULL
Booking_Status	varchar(15)	NO		'Confirmed'

```
DESCRIBE Payment;
```

Output:

Field	Type	Null	Key	Default
Payment_ID	varchar(10)	NO	PRI	NULL
Booking_ID	varchar(10)	NO	UNI	NULL
Amount	decimal(8,2)	NO		NULL
Payment_Date	date	NO		NULL
Payment_Method	varchar(10)	NO		'Cash'
Payment_Status	varchar(10)	NO		'Pending'

Verifying integrity: statements the database must reject

Test 1: duplicate primary key

```
INSERT INTO Customer VALUES
('CU01','Test User','9000000001','test1@mail.com','Salem');
```

Test 2: duplicate phone number (UNIQUE)

```
INSERT INTO Customer VALUES
('CU09','Test User','9876543210','test2@mail.com','Salem');
```

Test 3: invalid phone number (CHECK)

```
INSERT INTO Customer VALUES
('CU09','Test User','98765','test3@mail.com','Salem');
```

Test 4: invalid status value (CHECK)

```
INSERT INTO Booking VALUES
('B17','CU01','C01','2026-09-30','Mall','Airport',10,'Maybe');
```

Test 5: negative fare (CHECK)

```
INSERT INTO Car VALUES
('C08','KA07QR1111','Ritz','Hatchback',-5,'D02');
```

Test 6: pickup and drop at the same place (CHECK)

```
INSERT INTO Booking VALUES
('B17','CU01','C01','2026-09-30','Airport','Airport',10,'Confirmed');
```

Test 7: foreign key to a customer that does not exist

```
INSERT INTO Booking VALUES
('B17','CU99','C01','2026-09-30','Mall','Airport',10,'Confirmed');
```

Test 8: second payment for the same booking (UNIQUE)

```
INSERT INTO Payment VALUES
('P17','B01',300.00,'2026-09-07','Cash','Paid');
```

Test 9: deleting a customer who has bookings (ON DELETE RESTRICT)

```
DELETE FROM Customer WHERE Customer_ID = 'CU01';
```

Test 10: missing compulsory value (NOT NULL)

```
INSERT INTO Customer (Customer_ID, Phone) VALUES ('CU09','9000000009');
```

Verifying the DEFAULT constraints

```
INSERT INTO Customer (Customer_ID, Customer_Name, Phone)
VALUES ('CU09', 'Test Customer', '9000000009');
INSERT INTO Driver (Driver_ID, Driver_Name, Phone, License_No)
VALUES ('D07', 'Test Driver', '9000000010', 'DL1007');
INSERT INTO Booking (Booking_ID, Customer_ID, Car_ID, Pickup_Location,
                     Drop_Location, Distance_KM)
VALUES ('B99', 'CU09', 'C01', 'Mall', 'Airport', 10);
```

```
SELECT Customer_ID, Address FROM Customer WHERE Customer_ID = 'CU09';
```

Output:

Customer_ID	Address
CU09	Not Provided

```
SELECT Driver_ID, Availability FROM Driver WHERE Driver_ID = 'D07';
```

Output:

Driver_ID	Availability
D07	Available

```
SELECT Booking_ID, Booking_Status, Booking_Date IS NOT NULL AS Date_Filled FROM
Booking WHERE Booking_ID = 'B99';
```

Output:

Booking_ID	Booking_Status	Date_Filled
B99	Confirmed	1

```
DELETE FROM Booking  WHERE Booking_ID  = 'B99';
DELETE FROM Customer WHERE Customer_ID = 'CU09';
DELETE FROM Driver   WHERE Driver_ID   = 'D07';
```

Verifying the relationships (ON UPDATE CASCADE)

```
INSERT INTO Booking VALUES
('B98','CU08','C01','2026-09-30','Mall','Airport',10,'Confirmed');
UPDATE Customer SET Customer_ID = 'CU88' WHERE Customer_ID = 'CU08';
```

```
SELECT Booking_ID, Customer_ID FROM Booking WHERE Booking_ID = 'B98';
```

Output:

Booking_ID	Customer_ID
B98	CU88

```
UPDATE Customer SET Customer_ID = 'CU08' WHERE Customer_ID = 'CU88';
DELETE FROM Booking WHERE Booking_ID = 'B98';
```

Checking that no orphan records exist

```
SELECT b.Booking_ID FROM Booking b
LEFT JOIN Customer c ON b.Customer_ID = c.Customer_ID
WHERE c.Customer_ID IS NULL;
```

Output:

```
Empty set
```

```
SELECT p.Payment_ID FROM Payment p
LEFT JOIN Booking b ON p.Booking_ID = b.Booking_ID
WHERE b.Booking_ID IS NULL;
```

Output:

```
Empty set
```